PROJECT REPORT

MEDICINE STOCK MANAGEMENT SYSTEM

A Java-Based Web Application

Submitted by: G. Vamsi, P. Bhanu Prasad, M. Harika Sivani

Roll Numbers: 24ME1A0524, 24ME1A050, 24ME1A0543

Department: CSE

College: Ramachandra College of Engineering

1. Introduction

Medicine Stock Management System is a web-based application designed to help a pharmacy store room maintain medicine batch details, monitor stock quantities and track expiry dates. It provides a simple way to record stock additions and issues while preventing expired medicine batches from being issued.

2. Problem Statement

A pharmacy store room needs to monitor medicine batches, stock quantities and expiry dates. Manual tracking can make it difficult to know the available quantity of each medicine and identify expired batches. The system should maintain medicine details, record stock additions and reductions, display current stock and block the issue of expired medicine.

3. Objectives

- Maintain medicine name, batch number, quantity, price and expiry date.
- Record every stock addition and reduction.
- Display the available quantity for each medicine batch.
- Identify batches whose expiry date has passed.
- Prevent expired medicine from being issued.

4. Scope of the Project

The application is intended for a small pharmacy store room or an academic demonstration. It focuses on batch-level inventory tracking, stock movement recording, availability display and expiry validation. It does not replace a certified pharmacy or enterprise inventory platform.

5. Functional Requirements

Medicine management: The system shall store medicine name, batch number, quantity, price and expiry date for each batch.

Stock management: The system shall allow authorized users to add stock and issue stock, and shall record each stock movement.

Availability: The system shall display the current available quantity for every medicine batch.

Expiry management: The system shall identify a batch as expired when its expiry date is earlier than the current date.

Issuing control: The system shall not allow an expired batch to be issued. It shall also prevent issuing a quantity greater than available stock.

6. Non-Functional Requirements

- Usability: the interface should be simple and easy to understand.
- Reliability: stock quantities should be updated consistently.

- Performance: common operations should respond quickly for a small inventory.
- Maintainability: the code should be organized into separate application classes.
- Data persistence: medicine and transaction data should remain stored after restarting the application.

7. Technology Stack

- Programming language: Java 17
- Backend framework: Spring Boot
- Frontend: HTML, CSS and Thymeleaf templates
- Database: H2 embedded database
- Build tool: Maven
- Development environment: Visual Studio Code

8. System Modules

Medicine Batch Module: Adds and stores medicine batch information.

Stock Module: Updates quantity when stock is added or issued.

Availability Module: Shows current quantity for each batch.

Expiry Module: Compares expiry date with the current date and displays Expired or Active.

Issue Validation Module: Blocks expired batches and issues larger than the available quantity.

Transaction Module: Maintains a recent history of stock additions and issues.

9. System Workflow

1. Open the Medicine Stock Management website in a browser.
2. Enter medicine name, batch number, quantity, price and expiry date.
3. The system saves the batch and displays its current stock.
4. When new stock arrives, enter the quantity in the Add action.
5. When medicine is issued, enter the required quantity in the Issue action.
6. The system checks expiry status and available quantity before issuing.
7. The dashboard displays the updated quantity and recent stock transactions.

10. Database Design

Medicine table: id (primary key), name, batch_number, quantity, price and expiry_date. Each row represents one medicine batch.

Stock_Transaction table: id (primary key), medicine_id, medicine_name, type (ADDED or ISSUED), quantity and created_at. Each row records a stock movement.

11. Validation Rules

- Medicine name and batch number are required.

- Quantity and price cannot be negative.
- Stock additions and issue quantities must be greater than zero.
- An issue is rejected if the batch is expired.
- An issue is rejected if requested quantity exceeds available stock.

12. Expected Output

The website displays a medicine batch form, a medicine availability table with batch and expiry status, stock Add and Issue actions, and a recent transaction table. Successful stock operations update the available quantity. Expired batches are marked clearly and cannot be issued.

13. Advantages

- Reduces manual stock calculation.
- Makes batch-wise quantity visible.
- Helps staff notice expired stock.
- Maintains a basic record of stock movements.
- Prevents accidental issue of expired batches through application validation.

14. Limitations and Future Enhancements

This version is an educational prototype. Future versions can include user login and role-based access, barcode scanning, low-stock alerts, downloadable reports, supplier details, multiple store rooms, automated backups and a production-grade database.

15. Conclusion

The Medicine Stock Management System provides a simple digital method to maintain medicine batch details, track stock additions and reductions, view available quantities and identify expired batches. Its issue validation rule prevents expired batches from being issued through the application. The project demonstrates how Java and Spring Boot can be used to develop a basic inventory management website.

16. References

- Java Documentation: https://docs.oracle.com/en/java/
- Spring Boot Reference: https://docs.spring.io/spring-boot/index.html
- Thymeleaf Documentation: https://www.thymeleaf.org/documentation.html

17. Screenshots of the Website

The following screenshots show the Medicine Stock Management website running locally in the browser.

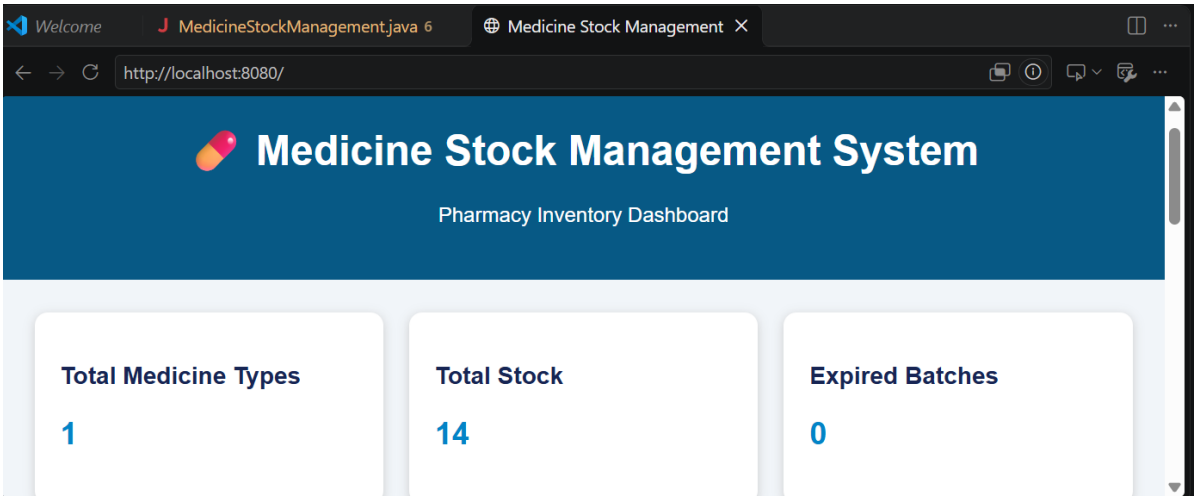


Figure 1: Medicine Stock Management dashboard showing summary cards.

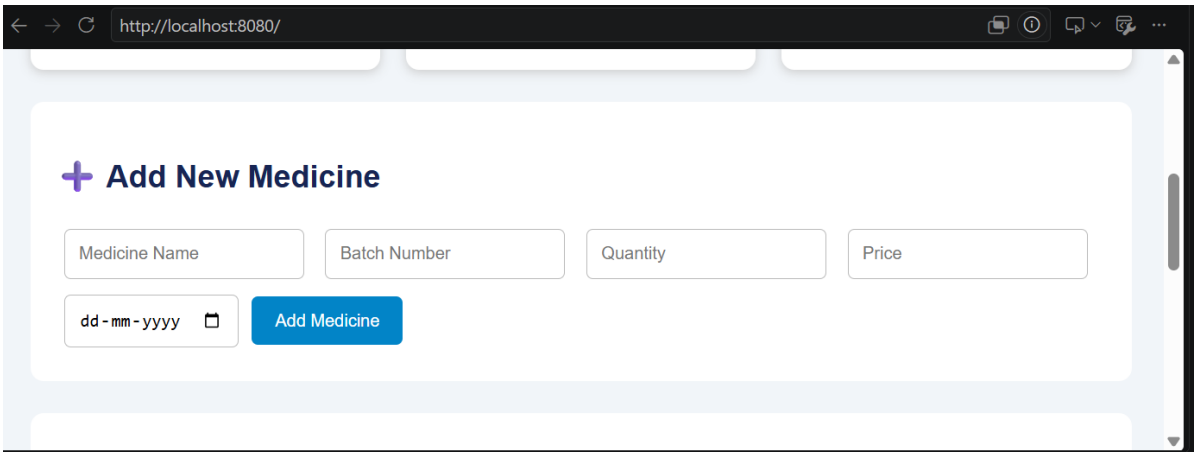


Figure 2: Add New Medicine form for entering medicine and batch details.

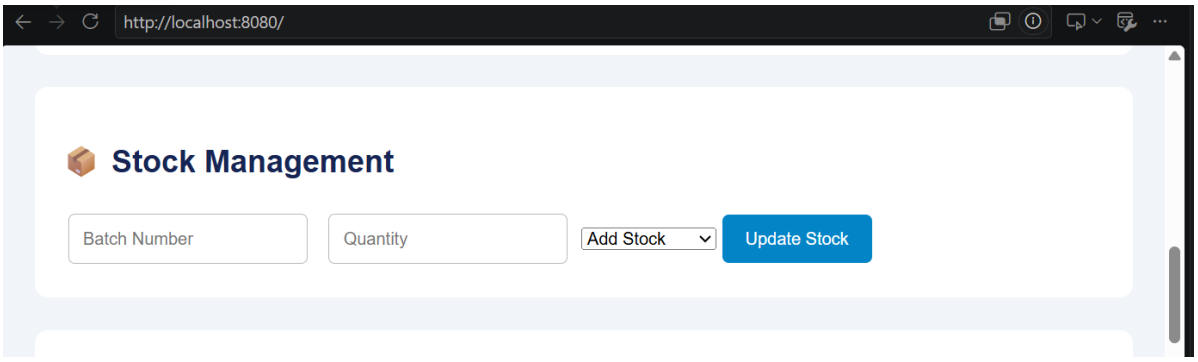


Figure 3: Stock Management section for adding or reducing stock.

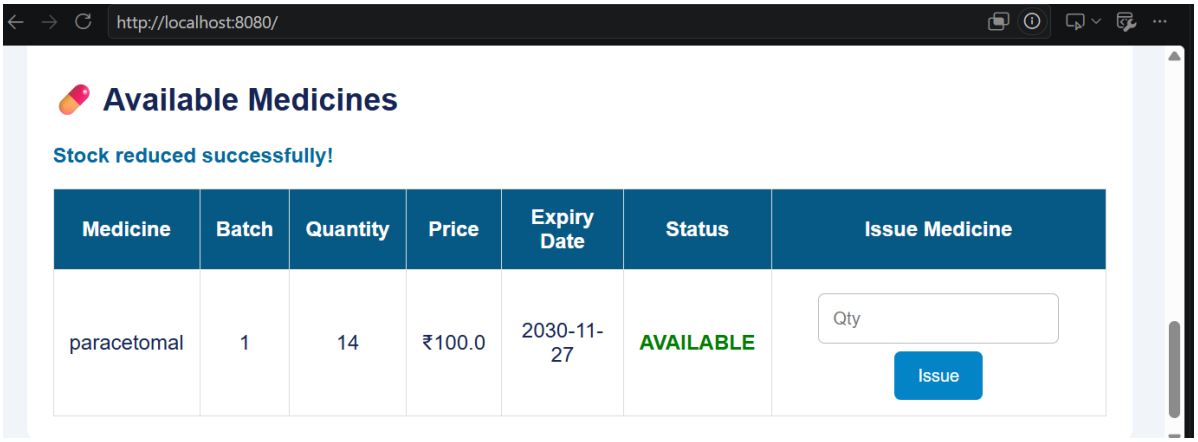


Figure 4: Available Medicines table showing quantity, price, expiry date and status.